

Counterfactual Sufficiency: State Representation and Response Geometry

Chae-Yeon Xon

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Abstract

Structural counterfactuals depend on the distinctions preserved by a state representation. A supplied map q and target family $\mathcal{T}$ induce exact and loss-relevant quotients. On a smooth fiber of q , weighted target derivatives form an operator whose rank gives the local coordinate count and whose spectral tail gives decoder loss in a shrinking neighborhood. A conditional generator bounds population decoder risk over a declared coordinate class. A finite-graph analogue covers discrete structural states, and cross-fitted coordinate selection satisfies an oracle bound. Across 20 fixed core-state splits of two KW97 parameterizations, minimum out-of-fold training risk selects the held-out oracle in all 80 three-coordinate comparisons. The median target-RMSE gain over the local spectrum is 4.29 percent, and every split has a positive four-comparison mean gain. A heterogeneous-agent transition economy illustrates the distribution-state construction.

JEL codes: C14, C52, C61, C63, D81. *Keywords:* counterfactual sufficiency, state representation, response geometry, dimension reduction, active subspaces, structural models.

1. Introduction

A structural model maps its state into choices, equilibrium transitions, and policy outcomes. Applied work often begins with a representation supplied by modeling or computation: moments of a distribution, accumulated human capital, a network summary, or a filtered latent state. That representation may merge distinctions that matter for policy targets. The target-relevant state content then lies within its fibers.

Consider two households with the same retained assets, income, and current choice. Differences in omitted experience or expectations may still generate different responses to a future subsidy. A state representation is sufficient for that target when every distinction it merges leaves the declared response vector unchanged. The criterion allows heterogeneous states to be compressed while preserving the responses used by the policy counterfactual.

Fix a supplied representation q and a family of structural counterfactuals $\mathcal{T}$. States sharing both q and the target vector form an exact counterfactual-equivalence class. A positive-semidefinite target loss produces a coarser, loss-relevant quotient. On a smooth fiber, the vertical space $\mathcal{V}_s = \ker Dq(s)$ collects the hidden state movements. Their weighted target derivatives define an operator $H_{\mathcal{T}}(s)$. Its rank gives the local coordinate count for first-order declared loss, and its spectral tail gives decoder risk in a shrinking neighborhood. Curvature bounds relate this local calculation to experiments with a finite radius.

The quality of an actual coordinate map reflects the local spectrum together with fiber geometry and the class of coordinates that can be measured or computed. For a declared class $\mathfrak{B}_d$, a conditional weighted generator decomposes decoder risk by frequency within each fiber and yields a population-risk certificate. A finite graph provides the corresponding construction for discrete structural states. The target family, loss, state distribution, state metric, tolerance, coordinate resolution, and coordinate class together form the state-design contract. Cross-fitted minimization of the certificate gives an oracle bound over finite-complexity classes. When response derivatives are observed only at scattered states, cross-fitted products and local reconstruction recover the local operator. Directional spectral inference then maps a loss tolerance into a confidence-protected dimension.

The construction draws on counterfactual equivalence, model reduction, active subspaces, Markov-generator spectral calculus, and goal-oriented design (Beraja 2023; Hebden and Winkler 2026; Sussmann 1976; Kotta, Moog, and Tönso 2018; Givan, Dean, and Greig 2003; Moore 1981; Constantine, Dow, and Wang 2014; Zahm et al. 2020; Bakry, Gentil, and Ledoux 2014). Equal-budget representations can be ranked differently by local derivative spectra and global decoder risk, as in the oscillatory approximation mechanism of Constantine et al. (2017). The generator certificate adds the within-fiber frequency information used for global selection.

Two quantitative exercises illustrate the distinction. A nonlinear incomplete-markets transition economy applies the construction to a distributional state and fiscal transition paths. A fixed KW97 model compares equal-budget coordinate families on held-out policy-response vectors and evaluates the finite-graph certificate on the same state-design contract. Across 20 core-state splits, direct out-of-fold risk selects the held-out oracle in all 80 model-run-split comparisons at the three-coordinate budget. The selected family reduces held-out target RMSE by 4.29 percent at the median, and the split-level mean gain ranges from 2.76 to 7.67 percent.

The target quotients lead from the local operator to the global decoder problem and the coordinate-selection certificate. Estimation, inference, and computation then connect the construction to the heterogeneous-agent and KW97 exercises.

2. State representations and counterfactual targets

2.1. Dynamic structural environment

Let $S_t \in \mathcal{S} \subseteq \mathbb{R}^K$ be a structural state, $A_t \in \mathcal{A}$ an action, $p \in \mathcal{P} \subseteq \mathbb{R}^q$ a policy environment, and $\theta \in \Theta \subseteq \mathbb{R}^p$ a structural parameter. A stationary infinite-horizon discrete-choice model can be written as

$$(1) \quad V_\theta(s, p) = \log \sum_{a \in \mathcal{A}} \exp \{ \pi_\theta(a, s, p) + \beta \mathbb{E}_\theta[V_\theta(S', p) \mid s, a, p] \},$$

where the log-sum form corresponds to independent type-I extreme-value choice shocks. The definitions below apply to structural models that generate a differentiable counterfactual map. A supplied representation is a C^1 map

$$(2) \quad q : \mathcal{S} \rightarrow \mathcal{X}.$$

The map can collect observed aggregates, administrative records, state variables retained by a computational approximation, or a previous-stage representation. Fix a state region on which q has constant rank. The vertical space is

$$(3) \quad \mathcal{V}_s = \ker Dq(s).$$

A Riemannian metric on $\mathcal{S}$ supplies units for state perturbations. Within one chart, let $B(s) : \mathbb{R}^k \rightarrow \mathcal{V}_s$ be an isometry, where $k = \dim \mathcal{V}_s$.

2.2. Counterfactual family

A target view is $t = (u, \delta, c, h)$, where u is a policy direction, δ is a policy displacement, c indexes an outcome selector C_c , and h indexes a horizon. Let p_0 be the baseline policy. The finite target response is

$$(4) \quad R_t(s; \theta) = C_c [T_h(s, p_0 + \delta u; \theta) - T_h(s, p_0; \theta)].$$

Let $\mathcal{T}$ be a finite family of views. Stack R_t into $\mathbf{R}_{\mathcal{T}}$. A continuous family can be treated by replacing sums with integrals.

DEFINITION 1 (State-design contract). *A state-design contract is*

$$(5) \quad \mathcal{E} = (q, \mathcal{T}, \{\omega_t, W_t\}_{t \in \mathcal{T}}, g, \mu, \varepsilon, \{\mathfrak{B}_d\}_{d \geq 0}),$$

where $W_t \geq 0$ is a declared target loss, g is the state metric, μ is the state distribution used for integrated risk, ε is a target-loss tolerance, and $\mathfrak{B}_d$ is the admissible class of d -coordinate augmentations. The contract is fixed before coordinate estimation and risk evaluation.

The exact quotient below depends on $(q, \mathcal{T})$. Local spectra additionally depend on the state and target metrics. Integrated risk and selected dimension additionally depend on $(\mu, \varepsilon, \mathfrak{B}_d)$. Reporting results under economically relevant alternative contracts gives a sensitivity frontier for the state conclusion.

DEFINITION 2 (Counterfactual equivalence). *For $s, s' \in \mathcal{S}$, write $s \sim_{\mathcal{T}, q} s'$ when $q(s) = q(s')$ and $\mathbf{R}_{\mathcal{T}}(s; \theta_0) = \mathbf{R}_{\mathcal{T}}(s'; \theta_0)$.*

The quotient $\mathcal{S}/\sim_{\mathcal{T}, q}$ records the distinctions in the structural state that are retained by q or used by the declared target family.

DEFINITION 3 (Loss-relevant counterfactual equivalence). *For $s, s' \in \mathcal{S}$, write $s \sim_{\mathcal{T}, W, q} s'$ when $q(s) = q(s')$ and $W_t^{1/2} R_t(s; \theta_0) = W_t^{1/2} R_t(s'; \theta_0)$ for every $t \in \mathcal{T}$.*

The loss-relevant quotient $\mathcal{Z}_W^* = \mathcal{S}/\sim_{\mathcal{T}, W, q}$ is a coarsening of the exact quotient. The two coincide when every W_t is positive definite on the span of attainable target differences. A singular decision loss retains the target combinations that affect the declared local decision problem.

THEOREM 1 (Counterfactual quotient and local repair). *Let $\pi^* : \mathcal{S} \rightarrow \mathcal{Z}^* = \mathcal{S}/\sim_{\mathcal{T}, q}$ be the quotient map. There are unique maps $\bar{q}$ and $\bar{\mathbf{R}}_{\mathcal{T}}$ on $\mathcal{Z}^*$ such that*

$$(6) \quad q = \bar{q} \circ \pi^*, \quad \mathbf{R}_{\mathcal{T}} = \bar{\mathbf{R}}_{\mathcal{T}} \circ \pi^*.$$

Let $z : \mathcal{S} \rightarrow \mathcal{Z}$ be any representation for which $q = q_z \circ z$ and $\mathbf{R}_{\mathcal{T}} = R_z \circ z$. Then there is a unique map $h : z(\mathcal{S}) \rightarrow \mathcal{Z}^*$ satisfying

$$(7) \quad \pi^* = h \circ z.$$

At a state s , define the stacked restricted derivative

$$(8) \quad A_{\mathcal{T}}(s) = D_s \mathbf{R}_{\mathcal{T}}(s; \theta_0) B(s).$$

Every C^1 augmentation $b : \mathcal{S} \rightarrow \mathbb{R}^d$ satisfying first-order target closure,

$$(9) \quad \ker D(q, b)(s) \subseteq \ker D\mathbf{R}_{\mathcal{T}}(s; \theta_0),$$

satisfies $d \geq \text{rank } A_{\mathcal{T}}(s)$. An augmentation of dimension $\text{rank } A_{\mathcal{T}}(s)$ attains first-order closure at s .

The first part of the theorem factors the targets through equivalence classes. The second turns target variation along the vertical space into a local coordinate count.

2.3. Policy-information operator

Give target view t a weight $\omega_t > 0$ and a target loss $W_t \geq 0$. Define

$$(10) \quad A_t(s) = D_s R_t(s; \theta_0) B(s)$$

and

$$(11) \quad H_{\mathcal{T}}(s) = \sum_{t \in \mathcal{T}} \omega_t A_t(s)' W_t A_t(s).$$

The weights specify the declared target loss. Sample size and design precision enter a separate operator in Section 3.

PROPOSITION 1 (Kernel of the target operator). *The operator in (11) satisfies*

$$(12) \quad \ker H_{\mathcal{T}}(s) = \bigcap_{t \in \mathcal{T}} \ker \{W_t^{1/2} A_t(s)\},$$

and

$$(13) \quad \text{rank } H_{\mathcal{T}}(s) = \text{rank } A_{\mathcal{T}, W}(s),$$

where $A_{\mathcal{T},W}$ stacks $\sqrt{\omega_t}W_t^{1/2}A_t$. If every W_t is positive definite on $\text{im } A_t(s)$, then $\ker H_{\mathcal{T}}(s) = \cap_t \ker A_t(s)$ and $\text{rank } H_{\mathcal{T}}(s) = \text{rank } A_{\mathcal{T}}(s)$. Under a change of vertical orthonormal frame $\tilde{B}(s) = B(s)O(s)$, $\tilde{H}_{\mathcal{T}}(s) = O(s)'H_{\mathcal{T}}(s)O(s)$.

The quotient tangent space can therefore be represented as

$$(14) \quad \mathcal{Q}_{\mathcal{T},W}(s) = \mathcal{V}_s / \ker H_{\mathcal{T}}(s).$$

REMARK 1 (Observability-Gramian special case). Consider a linearized state-output system $\delta s_{h+1} = F_h \delta s_h$ and $\delta y_h = C_h \delta s_h$, and let $\Phi_h = F_{h-1} \cdots F_0$ with $\Phi_0 = I$. If the declared targets are the outputs $\{\delta y_h\}_{h \in \mathcal{H}}$ and B is an orthonormal frame for the supplied-state vertical space, then

$$(15) \quad H_{\mathcal{T}} = B' \left\{ \sum_{h \in \mathcal{H}} \omega_h \Phi_h' C_h' W_h C_h \Phi_h \right\} B.$$

The matrix in braces is a weighted finite-horizon observability or output-sensitivity Gramian. Equation (15) therefore places the linear special case inside observability-based model reduction. Nonlinear empirical-Gramian methods construct related matrices from simulated or experimental state perturbations (Lall, Marsden, and Glavaski 2002; Hahn and Edgar 2002; Hahn, Edgar, and Marquardt 2003).

2.4. Metric covariance and distribution states

The operator in (11) is written in an orthonormal frame of the vertical space. Let $Z(s) : \mathbb{R}^k \rightarrow \mathcal{V}_s$ be an arbitrary local frame, let G_s be the coordinate matrix of the state metric restricted to $\mathcal{V}_s$, and define the target quadratic form

$$(16) \quad Q_{\mathcal{T},s} = \sum_{t \in \mathcal{T}} \omega_t \{D_s R_t(s) Z(s)\}' W_t \{D_s R_t(s) Z(s)\}.$$

The metric endomorphism $\mathcal{H}_{\mathcal{T},s}$ is characterized by

$$(17) \quad g_s(\mathcal{H}_{\mathcal{T},s} v, w) = \sum_{t \in \mathcal{T}} \omega_t \langle D_s R_t(s) v, D_s R_t(s) w \rangle_{W_t}, \quad v, w \in \mathcal{V}_s.$$

Its coordinate matrix is $G_s^{-1} Q_{\mathcal{T},s}$ and its spectral directions solve the generalized eigenproblem

$$(18) \quad Q_{\mathcal{T},s} v = \lambda G_s v.$$

PROPOSITION 2 (Coordinate covariance). Let $\tilde{z}$ be another vertical coordinate with nonsingular Jacobian C such that $z = C\tilde{z}$ at s . Then $\tilde{G}_s = C' G_s C$ and $\tilde{Q}_{\mathcal{T},s} = C' Q_{\mathcal{T},s} C$. The generalized eigenvalues in (18) are unchanged, and the corresponding tangent subspaces are carried by C .

A target-coordinate transformation $y = D\tilde{y}$ with nonsingular D preserves $Q_{\mathcal{T},s}$ when the target metric transforms as $\tilde{W} = D'WD$.

The state metric therefore enters the definition of the loss geometry. Units, normalization, and reparameterization are represented through the metric and its pullback. Grey (2026) develops an intrinsic Riemannian formulation of active subspaces with parallel transport across tangent spaces. For a distribution state μ in the interior of the probability simplex, the tangent space is $\{h : \mathbf{1}'h = 0\}$. The Fisher–Rao metric is

$$(19) \quad g_\mu(h, k) = \sum_{i=1}^n \frac{h_i k_i}{\mu_i}.$$

Chentsov’s characterization identifies this metric up to scale among Markov-invariant Riemannian metrics on finite probability simplices (Fujiwara 2024). Section 8 uses (19) for a distributional state.

2.5. Function-space target operator

Let $\mathbb{V}_s$ be a separable Hilbert vertical space with inner product $\langle \cdot, \cdot \rangle_s$. A target view t has a separable Hilbert output space $\mathbb{Y}_t$, a bounded positive operator W_t on $\mathbb{Y}_t$, and a bounded derivative $A_t : \mathbb{V}_s \rightarrow \mathbb{Y}_t$. Let ν be a finite measure on the target family and let ω_t be measurable and positive. Assume

$$(20) \quad \int_{\mathcal{T}} \omega_t \|W_t^{1/2} A_t\|_{\text{HS}}^2 d\nu(t) < \infty.$$

Define the target operator

$$(21) \quad \mathfrak{H}_{\mathcal{T},s} = \int_{\mathcal{T}} \omega_t A_t^* W_t A_t d\nu(t),$$

where the integral is a Bochner integral in the trace-class operators on $\mathbb{V}_s$.

THEOREM 2 (Hilbert target operator and sieve limit). *Under (20), $\mathfrak{H}_{\mathcal{T},s}$ is positive, self-adjoint, trace class, and compact, with*

$$(22) \quad \text{tr } \mathfrak{H}_{\mathcal{T},s} = \int_{\mathcal{T}} \omega_t \|W_t^{1/2} A_t\|_{\text{HS}}^2 d\nu(t).$$

If W_t is positive definite on the closure of $\text{im } A_t$ for ν -almost every t , then

$$(23) \quad \ker \mathfrak{H}_{\mathcal{T},s} = \{v \in \mathbb{V}_s : A_t v = 0 \text{ for } \nu\text{-almost every } t\}.$$

Let Π_m be finite-rank orthogonal projections with $\Pi_m v \rightarrow v$ for every $v \in \mathbb{V}_s$, and set $\mathfrak{H}_m = \Pi_m \mathfrak{H}_{\mathcal{T},s} \Pi_m$. Then

$$(24) \quad \|\mathfrak{H}_m - \mathfrak{H}_{\mathcal{T},s}\|_1 \rightarrow 0,$$

where $\|\cdot\|_1$ is the trace norm. If the nonzero eigenvalues are ordered as $\lambda_1 \geq \lambda_2 \geq \dots$, padded by zeros, then

$$(25) \quad \sum_{j \geq 1} |\lambda_j(\mathfrak{H}_m) - \lambda_j(\mathfrak{H}_{\mathcal{T},s})| \rightarrow 0.$$

A spectral gap at d yields convergence of the leading rank- d spectral projector in operator norm. The rank- d quadratic target loss equals $\sum_{j > d} \lambda_j$.

The direct-gradient special case belongs to infinite-dimensional active-subspace analysis (Kundu and Wycoff 2025). The target operator here acts on the vertical space induced by the supplied representation and stacks the declared structural counterfactual derivatives before the sieve approximation.

COROLLARY 1 (Distributional score representation). *Let μ be a positive density and let $\mathbb{H}_\mu = L_0^2(\mu)$ be the centered score Hilbert space with inner product $\langle f, g \rangle_\mu = \mathbb{E}_\mu[fg]$. Suppose the supplied representation records moments $\mathbb{E}_\mu[\phi_j]$, $j = 1, \dots, m$. The vertical score space is*

$$(26) \quad \mathbb{V}_\mu = \{s \in \mathbb{H}_\mu : \mathbb{E}_\mu[\phi_j s] = 0, j = 1, \dots, m\}.$$

If a scalar target derivative has Riesz representation $DR_t(\mu)[\mu s] = \mathbb{E}_\mu[\psi_t s]$, let $\psi_t^\vee$ be the orthogonal projection of ψ_t onto $\mathbb{V}_\mu$. Then

$$(27) \quad \mathfrak{H}_{\mathcal{T},\mu} s = \int_{\mathcal{T}} \omega_t \psi_t^\vee \mathbb{E}_\mu[\psi_t^\vee s] d\nu(t).$$

For a finite family of M scalarized target views, $\text{rank } \mathfrak{H}_{\mathcal{T},\mu} \leq M$. Its nonzero eigenvalues equal those of the $M \times M$ view Gram matrix with entries $\mathbb{E}_\mu[\psi_t^\vee \psi_{t'}^\vee]/M$ under equal weights.

The score map and Fisher pairing are standard constructions in nonparametric information geometry (Pistone and Sempì 1995; Cheng and Tong 2026). Corollary 1 gives a distributional implementation in which the operator dimension can be indexed by the target family while the distributional state remains function valued.

3. Counterfactual loss and information

3.1. Dimension budgets

Let the ordered eigenvalues of $H_{\mathcal{T}}(s)$ be $\lambda_1(s) \geq \dots \geq \lambda_k(s) \geq 0$. A rank- d orthogonal projector P retains d vertical coordinates. Define the quadratic first-order target loss

$$(28) \quad \mathcal{L}_s(P) = \text{tr}[(I - P)H_{\mathcal{T}}(s)].$$

THEOREM 3 (Target compression). *Among rank- d orthogonal projectors,*

$$(29) \quad \min_{\text{rank } P=d} \mathcal{L}_s(P) = \sum_{j>d} \lambda_j(s),$$

and a minimizer is the projector $P_d(s)$ onto the leading d eigenvectors of $H_{\mathcal{T}}(s)$. If vertical perturbations satisfy $\|v\|_2 \leq \rho$, the smallest worst-case first-order target norm over rank- d linear encoders equals

$$(30) \quad \rho \sqrt{\lambda_{d+1}(s)}.$$

For a tolerance $\varepsilon \geq 0$, the quadratic effective dimension is

$$(31) \quad r_\varepsilon(s) = \min \left\{ d : \sum_{j>d} \lambda_j(s) \leq \varepsilon \right\}.$$

3.2. Finite-radius representation frontier

Fix a supplied-fiber chart $\varphi_s : B_\rho^k \rightarrow q^{-1}(q(s))$ with $\varphi_s(0) = s$ and an isometric derivative $D\varphi_s(0) : \mathbb{R}^k \rightarrow \mathcal{V}_s$. Define the stacked weighted target increment

$$(32) \quad F_s(v) = \left\{ \sqrt{\omega_t} W_t^{1/2} [R_t(\varphi_s(v)) - R_t(s)] \right\}_{t \in \mathcal{T}}, \quad \|v\| \leq \rho.$$

Let $\mathcal{A}_s^\mathcal{T} = DF_s(0)$ and write its singular values as $\sigma_1 \geq \dots \geq \sigma_k$. For $d < k$, define

$$(33) \quad e_d(\rho) = \inf_{\substack{b: B_\rho^k \rightarrow \mathbb{R}^d \\ \text{continuous} \\ g}} \sup_{\|v\| \leq \rho} \|F_s(v) - g(b(v))\|.$$

THEOREM 4 (Continuous-encoder frontier). *Suppose*

$$(34) \quad F_s(v) = F_s(0) + \mathcal{A}_s^\mathcal{T} v + r_s(v), \quad \sup_{\|v\| \leq \rho} \|r_s(v)\| \leq K_s \rho^2 / 2.$$

Then

$$(35) \quad \max\{0, \rho\sigma_{d+1}(\mathcal{A}_s^\top) - K_s\rho^2/2\} \leq e_d(\rho) \leq \rho\sigma_{d+1}(\mathcal{A}_s^\top) + K_s\rho^2/2.$$

For the affine target $F_s(v) = F_s(0) + \mathcal{A}_s^\top v$,

$$(36) \quad e_d(\rho) = \rho\sigma_{d+1}(\mathcal{A}_s^\top).$$

The lower bound uses antipodal coincidence on the $(d+1)$ -dimensional leading right-singular sphere; the upper bound uses the leading right-singular coordinates.

3.3. Local prediction risk

Let U be supported on B_1^k and satisfy

$$(37) \quad \mathbb{E}U = 0, \quad \mathbb{E}UU' = cI_k, \quad \mathbb{E}[(I - P)U \mid PU] = 0$$

for every rank- d orthogonal projector P under comparison. Spherical perturbation laws satisfy (37). For $\rho > 0$, define the optimal local decoding risk

$$(38) \quad \mathfrak{R}_{s,\rho}(P) = \inf_h \mathbb{E} \left[\|F_s(\rho U) - h(PU)\|_2^2 \right].$$

THEOREM 5 (Local spectral-risk expansion). *Under (34), every rank- d orthogonal projector satisfies*

$$(39) \quad \left| \mathfrak{R}_{s,\rho}(P) - c\rho^2 \operatorname{tr}\{(I - P)H_{\mathcal{T}}(s)\} \right| \leq K_s\rho^3 \sqrt{c \operatorname{tr}\{(I - P)H_{\mathcal{T}}(s)\}} + \frac{K_s^2}{4}\rho^4.$$

Consequently,

$$(40) \quad \min_{\operatorname{rank} P=d} \mathfrak{R}_{s,\rho}(P) = c\rho^2 \sum_{j>d} \lambda_j(s) + O(\rho^3).$$

If $\lambda_d(s) - \lambda_{d+1}(s) = \gamma_d(s) > 0$ and $P_{s,\rho}$ minimizes (38), then

$$(41) \quad \|P_{s,\rho} - P_d(s)\|_F^2 = O\left(\frac{\rho}{\gamma_d(s)}\right).$$

The theorem gives the spectral tail a population prediction interpretation for perturbations that shrink within the supplied-state fiber. At a finite radius, held-out risk also depends on the perturbation law, target curvature, and decoder class.

3.4. Decision-induced target metric

Let a planner choose $a \in \mathbb{R}^m$ after forming a target vector $r \in \mathbb{R}^J$. The twice differentiable loss is $L(a, r)$. At (a_0, r_0) , suppose a_0 is an interior minimizer and $H_{aa} = D_{aa}L(a_0, r_0) > 0$.

THEOREM 6 (Decision metric). *Let $a^*(r)$ denote the local optimizer. Then*

$$(42) \quad Da^*(r_0) = -H_{aa}^{-1}D_{ar}L(a_0, r_0).$$

For a target perturbation Δr ,

$$(43) \quad L(a^*(r_0 + \Delta r), r_0) - L(a_0, r_0) = \frac{1}{2}\Delta r' W^{\text{dec}} \Delta r + o(\|\Delta r\|^2),$$

where

$$(44) \quad W^{\text{dec}} = D_{ra}L(a_0, r_0)H_{aa}^{-1}D_{ar}L(a_0, r_0) \geq 0.$$

The matrix W^{dec} can be used as the target metric in (11). It has rank at most m , so the decision-relevant quotient can merge target changes that leave the local optimal action unchanged. Decision-focused prediction uses downstream optimization loss to organize prediction tasks (Elmachtoub and Grigas 2022; Lee, Jin, and Lee 2026); equation (44) supplies the local metric used here. For a finite displacement v in the supplied-fiber chart used in (32), a twice differentiable target response satisfies

$$(45) \quad R_t(\varphi_s(v)) = R_t(s) + A_t(s)v + r_t(v), \quad \|r_t(v)\| \leq K_t\|v\|^2/2.$$

The first-order frontier therefore receives a remainder controlled by the declared state radius and target curvature. Policy magnitude can be included as a second expansion coordinate.

3.5. Coordinate realization

The projector $P_d(s)$ defines retained directions inside the vertical space. Let

$$(46) \quad \mathcal{K}_d(s) = B(s) \ker P_d(s) \subseteq \ker Dq(s)$$

be the discarded distribution.

PROPOSITION 3 (Local coordinate realization). *Suppose q and the constant-rank distribution $\mathcal{K}_d$ are smooth on a neighborhood of s_0 . After restricting the neighborhood, the following statements are equivalent:*

1. $\mathcal{K}_d$ is involutive;

2. there is a smooth map $b : U \rightarrow \mathbb{R}^d$ such that (q, b) has constant rank and

$$(47) \quad \ker D(q, b)(s) = \mathcal{K}_d(s), \quad s \in U.$$

The eigengap makes the spectral projector smooth when the target field has corresponding regularity. The eigengap leaves involutivity as a separate condition. A collection of local charts can cover several coordinate neighborhoods. A single global coordinate additionally requires a regular leaf space with a suitable global embedding.

3.6. Integrable-coordinate frontier

Let μ be a probability measure on $\mathcal{S}$. Let $b : \mathcal{S} \rightarrow \mathbb{R}^d$ be a smooth map such that $z = (q, b)$ has constant rank. In the vertical frame, let $P_b(s)$ be the rank- d projector onto the orthogonal complement of $\ker\{Db(s)|_{\mathcal{V}_s}\}$. Define the orientation loss

$$(48) \quad \Delta_b(s) = \text{tr}[\{P_d(s) - P_b(s)\}H_{\mathcal{T}}(s)].$$

PROPOSITION 4 (Integrable-coordinate loss). *Every admissible b satisfies*

$$(49) \quad \text{tr}\{(I - P_b(s))H_{\mathcal{T}}(s)\} = \sum_{j>d} \lambda_j(s) + \Delta_b(s), \quad \Delta_b(s) \geq 0.$$

When $\gamma_d(s) > 0$, $\Delta_b(s) = 0$ holds exactly when $P_b(s) = P_d(s)$. Moreover,

$$(50) \quad 0 \leq \Delta_b(s) \leq \|H_{\mathcal{T}}(s)\|_{\text{op}} \|P_b(s) - P_d(s)\|_*.$$

Hence

$$(51) \quad \inf_b \mathbb{E} \text{tr}\{(I - P_b(S))H_{\mathcal{T}}(S)\} \geq \mathbb{E} \sum_{j>d} \lambda_j(S).$$

Equality under a pointwise eigengap requires an admissible coordinate field aligned with the leading spectral field almost everywhere.

The spectral tail is a pointwise lower envelope. A smooth feature map must also have kernel fields that form level-set foliations. Once an admissible feature is supplied, equation (49) measures the loss created by this additional constraint.

3.7. Global decoding-risk certificate

Define the target map in its declared metric by

$$(52) \quad \mathbf{F}_{\mathcal{T}}(s) = \left\{ \sqrt{\omega_t} W_t^{1/2} R_t(s) \right\}_{t \in \mathcal{T}}$$

For a measurable feature z , define

$$(53) \quad \mathfrak{R}(z) = \inf_h \mathbb{E}_\mu \left[\|\mathbf{F}_T(S) - h(z(S))\|_2^2 \right].$$

The minimizing decoder is the conditional mean of $\mathbf{F}_T(S)$ given $z(S)$. Suppose a smooth submersion z admits conditional measures μ_ζ on its level sets $L_\zeta = z^{-1}(\zeta)$. Assume the scalar conditional Poincaré inequality

$$(54) \quad \int_{L_\zeta} |f - \mathbb{E}_{\mu_\zeta} f|^2 d\mu_\zeta \leq C(\zeta) \int_{L_\zeta} \|\nabla_{L_\zeta} f\|_g^2 d\mu_\zeta$$

for almost every ζ .

THEOREM 7 (Fiberwise target-risk certificate). *Suppose (54) holds, the coordinate restrictions of $\mathbf{F}_T$ belong to $H^1(L_\zeta, \mu_\zeta)$ almost everywhere, and the expression below is integrable. Then*

$$(55) \quad \mathfrak{R}(z) \leq \mathbb{E}_\mu \left[C(z(S)) \|\mathbf{D}\mathbf{F}_T(S) \Pi_{\ker D z(S)}\|_{\text{HS}}^2 \right].$$

For $z = (q, b)$,

$$(56) \quad \mathfrak{R}(q, b) \leq \mathbb{E}_\mu \left[C((q, b)(S)) \left\{ \sum_{j>d} \lambda_j(S) + \Delta_b(S) \right\} \right].$$

If $\ker D(q, b) = \mathcal{K}_d$ and $C \leq \bar{C}$ almost everywhere, then

$$(57) \quad \mathfrak{R}(q, b) \leq \bar{C} \mathbb{E}_\mu \sum_{j>d} \lambda_j(S).$$

The Poincaré constant depends on the geometry and conditional distribution of each feature fiber. Equation (56) separates spectral truncation from the directional loss created by an admissible coordinate map. The argument follows gradient-based ridge approximation and nonlinear level-set feature learning (Zahm et al. 2020; Verdière, Prieur, and Zahm 2025).

3.8. Disconnected and mixed state fibers

Suppose each regular level set of z is a countable union of connected components and let $C_z(S)$ be a measurable component label. Put

$$(58) \quad m(\zeta, c) = \mathbb{E}[\mathbf{F}_T(S) \mid z(S) = \zeta, C_z(S) = c], \quad \bar{m}(\zeta) = \mathbb{E}[\mathbf{F}_T(S) \mid z(S) = \zeta].$$

Assume a conditional Poincaré inequality on each component, with constant $C(\zeta, c)$.

THEOREM 8 (Mixed-fiber risk decomposition). *The optimal decoding risk satisfies*

$$(59) \quad \mathfrak{R}(z) \leq \mathbb{E} \left[C(z(S), C_z(S)) \|D\mathbf{F}_{\mathcal{T}}(S) \Pi_{\ker Dz(S)}\|_{\text{HS}}^2 \right]$$

$$(60) \quad + \mathbb{E} \left[\|m(z(S), C_z(S)) - \bar{m}(z(S))\|_2^2 \right].$$

Before the Poincaré step, the two terms form the exact conditional-variance decomposition into variation within connected components and variation between their conditional target means. Exact factorization through z holds when both terms vanish.

The second term in (60) detects target distinctions across latent types, regimes, and disconnected strata. Within-stratum derivatives supply the first term.

3.9. Admissible-coordinate selection

Fix before estimation a nonempty class $\mathfrak{B}_d$ of smooth maps $b : \mathcal{S} \rightarrow \mathbb{R}^d$ for which $z_b = (q, b)$ has constant rank and the conditions of Theorem 8 hold. The class encodes the measurement and implementation restrictions placed on a state representation. Define

$$(61) \quad \mathfrak{R}_d^* = \inf_{b \in \mathfrak{B}_d} \mathfrak{R}(q, b).$$

The global oracle problem therefore requires information that is absent from the pointwise spectrum.

THEOREM 9 (Local spectral concentration and global decoder risk). *For every integer $d \geq 1$, retained derivative share $\rho \in (0, 1)$, and risk ratio $K > 0$, there is a smooth scalar target on $[-1, 1]^{d+1}$ under the uniform distribution for which the leading d -dimensional eigenspace of the integrated derivative operator retains at least share ρ , while its optimal-decoder risk is at least K times the risk from another d -dimensional coordinate subspace.*

The oscillatory mechanism follows the ridge-approximation example in Constantine et al. (2017). Their calculation compares two coordinates at fixed frequencies and amplitudes. Theorem 9 selects these quantities jointly to attain arbitrary ρ and K and extends the construction to every retained dimension. The remainder of this subsection derives a global diagnostic from the slack in the fiberwise Poincaré inequality. For each connected component of a regular z_b -fiber, let $\mathcal{L}_{b, \zeta, c}$ be the nonnegative self-adjoint weighted Neumann generator associated with its conditional Dirichlet form. Assume compact resolvent and write $\nu_1(b, \zeta, c) > 0$ for its first nonzero eigenvalue. Put

$$(62) \quad r_b(S) = \mathbf{F}_{\mathcal{T}}(S) - m(z_b(S), C_{z_b}(S)), \quad \Gamma(b) = \mathbb{E} \|m(z_b(S), C_{z_b}(S)) - \bar{m}(z_b(S))\|_2^2.$$

The generator acts componentwise on r_b . Define the fiber certificate and its spectral-slack diagnostic by

$$(63) \quad \mathfrak{C}^\sharp(b) = \mathbb{E} \left[\nu_1(b, z_b(S), C_{z_b}(S))^{-1} \left\| D\mathbf{F}_T(S) \Pi_{\ker D z_b(S)} \right\|_{\text{HS}}^2 \right] + \Gamma(b),$$

$$(64) \quad \mathfrak{D}(b) = \mathbb{E} \left[\left\| \left\{ \nu_1(b, z_b(S), C_{z_b}(S))^{-1} \mathcal{L}_{b, z_b(S), C_{z_b}(S)} - I \right\} r_b \right\|_2^2 \right].$$

Write $\mathcal{A}_b r_b$ for the operator residual inside the norm in (64). For $z_b = (q, b)$, the derivative term in (63) equals the expectation of $\nu_1^{-1} \{ \sum_{j>d} \lambda_j + \Delta_b \}$. The next result replaces an unrestricted certificate gap by a target-specific quantity defined through the conditional generator.

THEOREM 10 (Fiber certificate sandwich). *Suppose the conditional generators above have compact resolvent, r_b lies in their domains, and the displayed quantities, including the squared norms of $\mathcal{L} r_b$, are integrable. Then every $b \in \mathfrak{B}_d$ satisfies*

$$(65) \quad \mathfrak{C}^\sharp(b) - \mathfrak{R}(q, b) = \mathbb{E} \langle r_b, \mathcal{A}_b r_b \rangle_2 \geq 0,$$

and

$$(66) \quad 0 \leq \mathfrak{C}^\sharp(b) - \mathfrak{R}(q, b) \leq \sqrt{\{\mathfrak{R}(q, b) - \Gamma(b)\} \mathfrak{D}(b)} \leq \sqrt{\mathfrak{R}(q, b) \mathfrak{D}(b)}.$$

$\mathfrak{C}^\sharp(b) = \mathfrak{R}(q, b)$ precisely when every nonzero conditional target residual lies in the first non-constant eigenspace of its connected fiber, up to mixtures across eigenfunctions sharing the first eigenvalue. If $\mathfrak{R}(q, b) > \Gamma(b)$, the first upper bound in (66) is attained precisely when $\mathcal{A}_b r_b = \kappa r_b$ in the direct-integral conditional L^2 space for one constant $\kappa \geq 0$. Residuals concentrated on one common normalized fiber frequency attain the bound.

Equation (65) is the conditional spectral decomposition of the Poincaré deficit; the spectral-gap and equality machinery is standard for symmetric Markov generators (Bakry, Gentil, and Ledoux 2014). Its role here is to supply the target- and coordinate-specific remainder in the risk oracle problem. The stated bounds use a positive conditional gap. Weak-Poincaré or other tail conditions provide the corresponding framework for zero-gap fibers. The $d = 1$ construction behind Theorem 9 gives an exact reading of the diagnostic. Let $\phi_k(u) = \cos\{k\pi(u+1)/2\}$ and $F(x, y) = \epsilon \phi_N(x) + \phi_1(y)$. On uniform interval fibers with the Neumann generator, $\mathfrak{D}(X) = 0$ and $\mathfrak{D}(Y) = \epsilon^2(N^2 - 1)^2/2$. The certificates are $\mathfrak{C}^\sharp(X) = 1/2$ and $\mathfrak{C}^\sharp(Y) = \epsilon^2 N^2/2$, while the risks are $1/2$ and $\epsilon^2/2$. At $\epsilon = 0.05$ and $N = 40$, the leading derivative direction retains 80 percent of integrated derivative trace and has 400 times the risk of the orthogonal coordinate. The risk-improving coordinate Y leaves a high-frequency residual, and $\mathfrak{D}(Y)$ records the resulting certificate slack exactly. For an affine target residual on a Gaussian fiber, the weighted generator is Ornstein–Uhlenbeck and $\mathfrak{D}(b) = 0$.

The diagnostic can be evaluated from target derivatives, conditional density scores, and second derivatives because on a Euclidean fiber with density p , $\mathcal{L}f = -p^{-1} \operatorname{div}(p \nabla f)$.

The same algebra has a finite-state implementation. Fix a declared resolution for the supplied coordinates, which partitions the evaluation states into cells $g \in \mathcal{G}$. For each candidate b , choose a connected reversible weighted graph within every cell; its edge rule can depend on distances in b , and its nonnegative self-adjoint Laplacian $L_{b,g}$ has first nonzero eigenvalue $\nu_{1,b,g}$. The cell resolution, edge rule, and bandwidth are part of the state-design contract. Let $r_{b,g}$ be a target residual from an externally fitted or cross-fitted decoder, let $\bar{r}_{b,g}$ be its cell mean, and put $r_{b,g}^\circ = r_{b,g} - \bar{r}_{b,g}$. Write $\|\cdot\|_g$ for the cell empirical norm and let π_g be the cell frequency.

PROPOSITION 5 (Finite-graph generator certificate). *Define*

$$(67) \quad \widehat{\mathfrak{R}}_G(b) = \sum_{g \in \mathcal{G}} \pi_g \left\{ \|r_{b,g}^\circ\|_g^2 + \|\bar{r}_{b,g}\|_2^2 \right\},$$

$$(68) \quad \widehat{\mathfrak{C}}_G(b) = \sum_{g \in \mathcal{G}} \pi_g \left\{ \nu_{1,b,g}^{-1} \langle r_{b,g}^\circ, L_{b,g} r_{b,g}^\circ \rangle_g + \|\bar{r}_{b,g}\|_2^2 \right\},$$

$$(69) \quad \widehat{\mathfrak{D}}_G(b) = \sum_{g \in \mathcal{G}} \pi_g \left\| \{ \nu_{1,b,g}^{-1} L_{b,g} - I \} r_{b,g}^\circ \right\|_g^2.$$

Then

$$(70) \quad 0 \leq \widehat{\mathfrak{C}}_G(b) - \widehat{\mathfrak{R}}_G(b) = \sum_{g \in \mathcal{G}} \pi_g \langle r_{b,g}^\circ, \{ \nu_{1,b,g}^{-1} L_{b,g} - I \} r_{b,g}^\circ \rangle_g$$

$$(71) \quad \leq \sqrt{\{ \widehat{\mathfrak{R}}_G(b) - \widehat{\Gamma}_G(b) \} \widehat{\mathfrak{D}}_G(b)},$$

where $\widehat{\Gamma}_G(b) = \sum_g \pi_g \|\bar{r}_{b,g}\|_2^2$. Conditional on an auxiliary sample used to construct a finite candidate menu, graphs, and decoders, suppose n_G independent evaluation clusters contribute scores in $[0, M_G]$ to (68). For a menu $\mathfrak{B}_{d,G}$, with probability at least $1 - \alpha$,

$$(72) \quad \sup_{b \in \mathfrak{B}_{d,G}} |\widehat{\mathfrak{C}}_G(b) - \mathbb{E} \widehat{\mathfrak{C}}_G(b)| \leq M_G \sqrt{\frac{\log\{2|\mathfrak{B}_{d,G}|/\alpha\}}{2n_G}}.$$

Proposition 5 applies graph-Laplacian spectral calculus (Chung 1997) and replaces conditional density-score and boundary estimation by a declared graph on a discrete structural state grid. The graph certificate remains an upper bound for the cross-fitted residual risk on that grid. Its empirical role is candidate selection within the declared graph, coordinate resolution, decoder, and target metric. A three-way split can separate candidate construction, graph and decoder fitting, and evaluation when sampling inference is required.

The statistical bound uses a declared finite-complexity coordinate class. Let $\mathfrak{B}_d = \{b_\eta : \eta \in \Xi_d\}$, where Ξ_d lies in a Euclidean ball of radius D_Ξ in $\mathbb{R}^{p_d}$. Let $\psi_\eta(S)$ denote the nonnegative observation-level score in (63), including the between-component term, so that $\mathfrak{C}^\sharp(b_\eta) = \mathbb{E}\psi_\eta(S)$. Assume

$$(73) \quad 0 \leq \psi_\eta(S) \leq M, \quad |\psi_\eta(S) - \psi_{\eta'}(S)| \leq L\|\eta - \eta'\|_2$$

almost surely. Cross-fitted first stages produce scores $\widehat{\psi}_{\eta,i}$ and

$$(74) \quad \widehat{\mathfrak{C}}_n(\eta) = \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\eta,i}, \quad \rho_n = \sup_{\eta \in \Xi_d} \left| \frac{1}{n} \sum_{i=1}^n \{\widehat{\psi}_{\eta,i} - \psi_\eta(S_i)\} \right|.$$

For $\alpha \in (0, 1)$, define

$$(75) \quad a_n(\alpha) = \frac{M}{\sqrt{n}} + M \sqrt{\frac{p_d \log\{1 + 4LD_\Xi\sqrt{n}/M\} + \log(2/\alpha)}{2n}}.$$

THEOREM 11 (Feasible-coordinate oracle bound). *For an optimization tolerance $\eta_n \geq 0$, let $\widehat{\eta}_d$ satisfy*

$$(76) \quad \widehat{\mathfrak{C}}_n(\widehat{\eta}_d) \leq \inf_{\eta \in \Xi_d} \widehat{\mathfrak{C}}_n(\eta) + \eta_n.$$

Suppose $\eta_d^* \in \Xi_d$ exists such that $b_d^* = b_{\eta_d^*}$ attains (61). Under (73), with probability at least $1 - \alpha$,

$$(77) \quad 0 \leq \mathfrak{R}(q, b_{\widehat{\eta}_d}) - \mathfrak{R}_d^* \leq \sqrt{\{\mathfrak{R}_d^* - \Gamma(b_d^*)\} \mathfrak{D}(b_d^*)} + 2\rho_n + 2a_n(\alpha) + \eta_n.$$

If $\mathfrak{D}(b_d^*) \leq \delta_d^2 \{\mathfrak{R}_d^* - \Gamma(b_d^*)\}$, the first term is at most $\delta_d \mathfrak{R}_d^*$. Writing $\overline{\mathfrak{D}}_d = \sup_{\eta \in \Xi_d} \mathfrak{D}(b_\eta)$, the same term is bounded by $\sqrt{M\overline{\mathfrak{D}}_d}$ because $\mathfrak{R}_d^* \leq M$.

COROLLARY 2 (Reported oracle bound). *Suppose deterministic sequences $\bar{\rho}_n, s_n \geq 0$ and a nonnegative estimator $\widehat{\mathfrak{D}}_n$ satisfy*

$$(78) \quad \Pr(\rho_n \leq \bar{\rho}_n) \geq 1 - \beta_{1n}, \quad \Pr\left(\sup_{\eta \in \Xi_d} |\widehat{\mathfrak{D}}_n(b_\eta) - \mathfrak{D}(b_\eta)| \leq s_n\right) \geq 1 - \beta_{2n}.$$

Then, with probability at least $1 - \alpha - \beta_{1n} - \beta_{2n}$,

$$(79) \quad \mathfrak{R}(q, b_{\widehat{\eta}_d}) - \mathfrak{R}_d^* \leq \sqrt{M \left\{ \sup_{\eta \in \Xi_d} \widehat{\mathfrak{D}}_n(b_\eta) + s_n \right\}} + 2\bar{\rho}_n + 2a_n(\alpha) + \eta_n.$$

Under the relative-slack condition in Theorem 11, with probability at least $1 - \alpha - \beta_{1n}$,

$$(80) \quad \mathfrak{R}(q, b_{\widehat{\eta}_d}) \leq (1 + \delta_d) \mathfrak{R}_d^* + 2\bar{\rho}_n + 2a_n(\alpha) + \eta_n.$$

A uniform rate r_n for all first-stage objects entering the score, including the target derivative, component means, and conditional spectral gap, gives $\bar{\rho}_n = O(r_n)$ on an event whose probability tends to one when the score functional is uniformly Lipschitz in those objects and the gap stays bounded away from zero. Evaluation of $\widehat{\mathfrak{D}}_n$ additionally uses the conditional density score and target second derivatives. If LD_{Ξ}/M grows at most polynomially in n , then $r_n = o(1)$, $p_d \log n/n = o(1)$, and $\eta_n = o_p(1)$ make the estimation and optimization terms vanish. The additive generator term also requires $\overline{\mathfrak{D}}_d = o(1)$, or its reported counterpart $\sup_{\eta} \widehat{\mathfrak{D}}_n(b_{\eta}) + s_n = o_p(1)$. These conditions state the estimation and approximation burdens created by the generator diagnostic and complete the population sandwich with sampling and approximation requirements. A constructive class can be defined on a declared q -adapted atlas. Within a chart (x, u) satisfying $q(x, u) = x$, let $\Psi_{\eta}(x, \cdot) : \mathbb{R}^k \rightarrow \mathbb{R}^k$ be a smoothly indexed diffeomorphism and set $b_{\eta}(x, u) = \pi_d \Psi_{\eta}(x, u)$. Compatibility restrictions on chart overlaps give a global feature. Affine maps, prespecified moment dictionaries, and fixed-architecture coupling flows provide finite-dimensional choices for Ξ_d . Equation (77) separates fiber-frequency slack, first-stage error, class complexity, and numerical optimization.

PROPOSITION 6 (Finite-state coordinate count). *Let $\mathcal{S}$ be finite. If q fails to carry the exact target quotient, there is a scalar map $b : \mathcal{S} \rightarrow \mathbb{R}$ such that (q, b) carries it exactly.*

One unrestricted real coordinate can label every class of a finite quotient. Complexity in a discrete representation must therefore be defined through a declared cost, such as quotient cardinality, code length, a measurement dictionary, or regularity on a supplied graph.

3.10. State measurement completeness

Let $M_{\mathcal{D}} : \mathcal{S} \rightarrow \mathbb{R}^r$ collect the state fields supplied by a measurement system, including q when the supplied representation is observed. Let $\mathbf{R}_{\mathcal{T}}$ stack the declared counterfactual targets. Work in an orthonormal frame of the supplied-state vertical space and let $P_{\mathcal{D}}(s)$ project onto $\ker DM_{\mathcal{D}}(s)$. Define the residual target operator

$$(81) \quad H_{\mathcal{T}|\mathcal{D}}(s) = P_{\mathcal{D}}(s)H_{\mathcal{T}}(s)P_{\mathcal{D}}(s).$$

THEOREM 12 (Counterfactual completeness and measurement repair). *At a state s , define*

$$(82) \quad \delta_{\mathcal{T}|\mathcal{D}}(s) = \text{rank}\left(D\mathbf{R}_{\mathcal{T}}(s)|_{\ker DM_{\mathcal{D}}(s)}\right) = \text{rank } D(M_{\mathcal{D}}, \mathbf{R}_{\mathcal{T}})(s) - \text{rank } DM_{\mathcal{D}}(s).$$

First-order target recovery from $M_{\mathcal{D}}$ holds exactly when $\delta_{\mathcal{T}|\mathcal{D}}(s) = 0$. Every collection of k added C^1 scalar measurements that achieves first-order recovery satisfies $k \geq \delta_{\mathcal{T}|\mathcal{D}}(s)$, and a collection with equality exists at the derivative level. Let $A_{\mathcal{T},W,\mathcal{D}}(s)$ stack $\sqrt{\omega_t}W_t^{1/2}DR_t(s)P_{\mathcal{D}}(s)$ and define the loss deficiency $\delta_{\mathcal{T},W|\mathcal{D}}(s) = \text{rank } A_{\mathcal{T},W,\mathcal{D}}(s)$. Then

$$(83) \quad \text{rank } H_{\mathcal{T}|\mathcal{D}}(s) = \delta_{\mathcal{T},W|\mathcal{D}}(s), \quad H_{\mathcal{T}|\mathcal{D}}(s) = 0 \iff \delta_{\mathcal{T},W|\mathcal{D}}(s) = 0.$$

Every collection of k added scalar measurements that eliminates first-order declared loss satisfies $k \geq \delta_{\mathcal{T},W|\mathcal{D}}(s)$, and equality is attainable at the derivative level. If the target metric is positive definite on the attainable residual target image, then $\delta_{\mathcal{T},W|\mathcal{D}}(s) = \delta_{\mathcal{T}|\mathcal{D}}(s)$. Let $\mu_1 \geq \mu_2 \geq \dots$ be the eigenvalues of $H_{\mathcal{T}|\mathcal{D}}(s)$. Among d additional orthonormal local scalar measurements with derivatives in the unresolved data fiber, the minimum remaining isotropic quadratic target loss is

$$(84) \quad \sum_{j>d} \mu_j,$$

attained by measuring the first d eigen-directions of $H_{\mathcal{T}|\mathcal{D}}(s)$. Suppose $M_{\mathcal{D}}$ and $(M_{\mathcal{D}}, \mathbf{R}_{\mathcal{T}})$ have constant ranks on a neighborhood with connected local fibers of $M_{\mathcal{D}}$. If $\delta_{\mathcal{T}|\mathcal{D}} = 0$ throughout that neighborhood, then there is a local map g such that $\mathbf{R}_{\mathcal{T}} = g \circ M_{\mathcal{D}}$.

The raw deficiency counts repair for exact target recovery. The residual operator measures declared-loss variation inside the state distinctions left unresolved by the measurement system; its rank counts loss repair and its spectrum orders added measurements under a dimension budget.

3.11. Observed design views

Let $\theta_0 \in \Theta \subseteq \mathbb{R}^p$. An observed design view $e \in \mathcal{D}$ supplies a moment vector $m_e(\theta) \in \mathbb{R}^{r_e}$ with

$$(85) \quad J_e = D_{\theta} m_e(\theta_0).$$

Stack active views to form $J_{\mathcal{D}}$. With $n_e > 0$ replications and moment covariance $\Sigma_e > 0$, define

$$(86) \quad I_{\mathcal{D}} = \sum_{e \in \mathcal{D}} n_e J_e' \Sigma_e^{-1} J_e.$$

Let $\tau(\theta) \in \mathbb{R}^m$ be a structural target and write $T_{\theta} = D_{\theta} \tau(\theta_0)$.

THEOREM 13 (Design information and target identification). *The information kernel is*

$$(87) \quad \ker I_{\mathcal{D}} = \bigcap_{e \in \mathcal{D}: n_e > 0} \ker J_e = \ker J_{\mathcal{D}}.$$

The target is locally identified to first order from the design when

$$(88) \quad \ker J_{\mathcal{D}} \subseteq \ker T_{\theta}.$$

Condition (88) is equivalent to a linear factorization $T_{\theta} = LJ_{\mathcal{D}}$ for a map L on the image of $J_{\mathcal{D}}$. The residual target-identification dimension is

$$(89) \quad r_{\mathcal{D} \rightarrow \tau} = \text{rank}(T_{\theta}|_{\ker J_{\mathcal{D}}}) = \text{rank} \begin{pmatrix} J_{\mathcal{D}} \\ T_{\theta} \end{pmatrix} - \text{rank} J_{\mathcal{D}}.$$

Every collection of unrestricted scalar added moments that yields (88) has cardinality at least $r_{\mathcal{D} \rightarrow \tau}$, and a collection of that cardinality exists at the derivative level.

A positive change in n_e scales one summand of (86) and preserves the kernel in (87). Adding a design view e^* changes the kernel to $\ker I_{\mathcal{D}} \cap \ker J_{e^*}$. Replication therefore changes information eigenvalues while additional views can change information rank.

COROLLARY 3 (Information accounting). *Let S be a block of additional design moments. Then*

$$(90) \quad r_{\mathcal{D} \rightarrow \tau} - r_{\mathcal{D} \cup S \rightarrow \tau} = \text{rank} J_{\mathcal{D} \cup S} + \text{rank} \begin{pmatrix} J_{\mathcal{D}} \\ T_{\theta} \end{pmatrix} - \text{rank} J_{\mathcal{D}} - \text{rank} \begin{pmatrix} J_{\mathcal{D} \cup S} \\ T_{\theta} \end{pmatrix} \geq 0.$$

3.12. Target-directed view allocation

Let $M > 0$ be the current precision matrix on a local state or parameter coordinate and let $\Sigma = M^{-1}$. Let $H_{\mathcal{T}} \geq 0$ measure the declared target loss. A candidate view has derivative $J \in \mathbb{R}^{r \times k}$ and noise covariance $R > 0$, giving the precision update $M^+ = M + J'R^{-1}J$. Define the target posterior-variance criterion

$$(91) \quad \Psi(M) = \text{tr}(H_{\mathcal{T}}M^{-1}).$$

PROPOSITION 7 (Target loss from an added view). *The reduction in (91) from the candidate view is*

$$(92) \quad \Psi(M) - \Psi(M^+) = \text{tr} \left[H_{\mathcal{T}} \Sigma J' (R + J \Sigma J')^{-1} J \Sigma \right].$$

For a scalar view with precision contribution $\nu aa'$,

$$(93) \quad \Psi(M) - \Psi(M + \nu aa') = \frac{\nu a' \Sigma H_{\mathcal{T}} \Sigma a}{1 + \nu a' \Sigma a}.$$

A sequential target-directed design selects the candidate view with the largest value in (92). The criterion is a local goal-oriented A-optimal design calculation (Attia, Alexanderian, and Saibaba 2018).

4. Scattered reconstruction

4.1. Local response estimates

Work on a compact state region $\mathcal{S}_0$ contained in one chart. The vertical dimension k is constant. A finite target family $\mathcal{T}$ is used in the main results. Let $s_{1n}, \dots, s_{N_n n}$ be anchor states. For each anchor and target view, obtain two cross-fitted response-derivative estimates $\widehat{A}_{it}^{(1)}$ and $\widehat{A}_{it}^{(2)}$. State-dependent response estimators can supply these local derivatives after their state and shock-size conditions are matched to the declared target; Gonçalves et al. (2024) give such conditions for local projections.

ASSUMPTION 1 (Response field and cover). *The region $\mathcal{S}_0$ is compact. The map q is C^1 with constant rank on a neighborhood of $\mathcal{S}_0$. The operators $A_t(s)$ are bounded uniformly over s and t and satisfy $\|A_t(s) - A_t(s')\|_{\text{op}} \leq L\|s - s'\|^\beta$ for some $\beta \in (0, 1]$. The weights and target metrics are bounded. Partition weights satisfy $\kappa_{ni}(s) \geq 0$, $\sum_i \kappa_{ni}(s) = 1$, and $\kappa_{ni}(s) > 0$ only when $\|s - s_{in}\| \leq h_n$, where $h_n \rightarrow 0$. The local estimates satisfy*

$$(94) \quad \max_{i,t,b \in \{1,2\}} \|\widehat{A}_{it}^{(b)} - A_t(s_{in})\|_{\text{op}} = O_p(\zeta_n), \quad \zeta_n \rightarrow 0.$$

Define the cross-fitted local patch

$$(95) \quad \widetilde{H}_i = \sum_{t \in \mathcal{T}} \frac{\omega_t}{2} \left[\widehat{A}_{it}^{(1)'} W_t \widehat{A}_{it}^{(2)} + \widehat{A}_{it}^{(2)'} W_t \widehat{A}_{it}^{(1)} \right]$$

and the reconstructed field

$$(96) \quad \widehat{H}_n(s) = \sum_{i=1}^{N_n} \kappa_{ni}(s) \widetilde{H}_i.$$

LEMMA 1 (Quadratic patch centering). *Suppose $\widehat{A}^{(b)} = A + E^{(b)}$ for $b \in \{1, 2\}$, with $\mathbb{E}[E^{(b)} | s] = 0$ and $\mathbb{E}[E^{(1)'} W E^{(2)} | s] = 0$. Then*

$$(97) \quad \mathbb{E} \left[\frac{1}{2} \{ \widehat{A}^{(1)'} W \widehat{A}^{(2)} + \widehat{A}^{(2)'} W \widehat{A}^{(1)} \} \middle| s \right] = A' W A.$$

For a same-sample estimate $\widehat{A} = A + E$ with $\mathbb{E}[E \mid s] = 0$,

$$(98) \quad \mathbb{E}[\widehat{A}' W \widehat{A} \mid s] = A' W A + \mathbb{E}[E' W E \mid s].$$

The construction can use repeated folds and average the cross products.

THEOREM 14 (Scattered reconstruction). *Under Assumption 1,*

$$(99) \quad \sup_{s \in \mathcal{S}_0} \|\widehat{H}_n(s) - H_{\mathcal{T}}(s)\|_{\text{op}} = O_p(r_n), \quad r_n = \zeta_n + \zeta_n^2 + h_n^\beta.$$

A numerical integration of a continuous target family adds its uniform quadrature error to r_n .

4.2. Frame-aligned reconstruction

Suppose each anchor uses a local vertical frame. Let $O_{i \rightarrow s}$ denote the orthogonal coordinate transport from the frame at s_i to the frame at s , and let $\widehat{O}_{i \rightarrow s}$ be an estimate. Define

$$(100) \quad \widehat{H}_n^A(s) = \sum_i \kappa_{ni}(s) \widehat{O}_{i \rightarrow s} \widetilde{H}_i \widehat{O}_{i \rightarrow s}'.$$

COROLLARY 4 (Aligned local frames). *Suppose Theorem 14's local patch rate holds in each anchor frame, $\sup_{i,s: \kappa_{ni}(s) > 0} \|\widehat{O}_{i \rightarrow s} - O_{i \rightarrow s}\|_{\text{op}} = O_p(\eta_n)$, and*

$$(101) \quad \sup_{i,s: \kappa_{ni}(s) > 0} \|O_{i \rightarrow s} H_{\mathcal{T}}(s_i) O_{i \rightarrow s}' - H_{\mathcal{T}}(s)\|_{\text{op}} = O(h_n^\beta).$$

Then

$$(102) \quad \sup_{s \in \mathcal{S}_0} \|\widehat{H}_n^A(s) - H_{\mathcal{T}}(s)\|_{\text{op}} = O_p(\zeta_n + \zeta_n^2 + h_n^\beta + \eta_n).$$

Local PCA and frame synchronization provide one route to the transport matrices in point-cloud settings (Singer and Wu 2012). In an observed Euclidean state with a common coordinate system, $O_{i \rightarrow s} = I$. A finite-sample positive-semidefinite version replaces $\widehat{H}$ by its positive spectral part $\widehat{H}_+$. Since $H_{\mathcal{T}} \geq 0$,

$$(103) \quad \|\widehat{H}_+ - H_{\mathcal{T}}\|_{\text{op}} \leq 2\|\widehat{H} - H_{\mathcal{T}}\|_{\text{op}},$$

so the reconstruction rate is preserved.

4.3. Eigenspaces and loss

ASSUMPTION 2 (Eigenspace separation). *For a retained dimension d ,*

$$(104) \quad \inf_{s \in \mathcal{S}_0} \{\lambda_d(s) - \lambda_{d+1}(s)\} \geq \gamma_d > 0.$$

Let $\widehat{P}_d(s)$ be the leading rank- d projector of $\widehat{H}_n(s)$.

COROLLARY 5 (Synthetic-state recovery). *Under Theorem 14 and Assumption 2,*

$$(105) \quad \sup_{s,j} |\widehat{\lambda}_j(s) - \lambda_j(s)| = O_P(r_n)$$

and

$$(106) \quad \sup_s \|\widehat{P}_d(s) - P_d(s)\|_{\text{op}} = O_P(r_n/\gamma_d).$$

On events satisfying $\sup_s \|\widehat{H}_n(s) - H_{\mathcal{T}}(s)\|_{\text{op}} < \gamma_d/2$, the target-loss regret satisfies

$$(107) \quad 0 \leq \mathcal{L}_s(\widehat{P}_d) - \mathcal{L}_s(P_d) \leq \frac{4d}{\gamma_d} \|\widehat{H}_n(s) - H_{\mathcal{T}}(s)\|_{\text{op}}^2.$$

Hence the uniform loss regret is $O_P(dr_n^2/\gamma_d)$.

If the target rank is constant on $\mathcal{S}_0$, $\inf_s \lambda_r(s) > 0$, and $\lambda_{r+1}(s) = 0$, a threshold $a_n \rightarrow 0$ with $r_n/a_n \rightarrow 0$ yields uniform rank recovery through $\widehat{r}(s) = \#\{j : \widehat{\lambda}_j(s) > a_n\}$.

5. Structural estimation

5.1. Criterion

Let $Q_B(\theta)$ denote a population criterion based on observed behavior, transitions, or equilibrium moments. The structural model generates a target field $H_{\mathcal{T},\theta}(s)$. The population criterion is

$$(108) \quad Q(\theta) = Q_B(\theta) + \lambda \int_{\mathcal{S}_0} \|H_{\mathcal{T},\theta}(s) - H_0(s)\|_F^2 d\nu(s),$$

where H_0 is the population field recovered from response data. When target responses are generated entirely by the structural model, the field can be computed after behavioral estimation; equation (108) applies when response information enters estimation.

5.2. Response-data field and model field

At a fixed state, let $Z_i \in \mathbb{R}^{k_m}$ be coordinates of an observed local perturbation in a sieve of the vertical space. For target view t , consider the local response equation

$$(109) \quad Y_{it} = \alpha_t + \beta_t' Z_i + \varepsilon_{it}, \quad \mathbb{E}[\varepsilon_{it} | Z_i] = 0.$$

The first stage estimates the response derivatives β_t from perturbation coordinates and target responses. Split-sample estimates $\widehat{\beta}_t^{(1)}$ and $\widehat{\beta}_t^{(2)}$ produce the cross-fitted field

$$(110) \quad \widehat{H}_m^D = \sum_t \frac{\omega_t}{2} \left\{ \widehat{\beta}_t^{(1)} \widehat{\beta}_t^{(2)'} + \widehat{\beta}_t^{(2)} \widehat{\beta}_t^{(1)'} \right\}.$$

A structural model indexed by θ generates $H_m^M(\theta)$ on the same sieve and state metric. One operator-matching criterion is

$$(111) \quad Q_m^H(\theta) = \|H_m^M(\theta) - H_m^D\|_{\Omega_m}^2.$$

The structural criterion in (112) allows (111) as the field component. The sieve result in Theorem 2 supplies the approximation link when the state is function valued. Let $\widetilde{H}_{\mathcal{T},\theta,n}$ be the numerically computed model field and let ν_n be the sample integration measure. The sample criterion is

$$(112) \quad \widehat{Q}_n(\theta) = \widehat{Q}_{B,n}(\theta) + \lambda \int_{\mathcal{S}_0} \|\widetilde{H}_{\mathcal{T},\theta,n}(s) - \widehat{H}_n(s)\|_F^2 d\nu_n(s).$$

ASSUMPTION 3 (Structural criterion). *The parameter space Θ is compact. The map $(\theta, s) \mapsto H_{\mathcal{T},\theta}(s)$ is jointly continuous and uniformly bounded. The behavioral criterion converges uniformly, $\sup_{\theta} |\widehat{Q}_{B,n}(\theta) - Q_B(\theta)| = o_p(1)$. The structural solver satisfies $\sup_{\theta,s} \|\widetilde{H}_{\mathcal{T},\theta,n}(s) - H_{\mathcal{T},\theta}(s)\|_F = o_p(1)$. The measure ν_n obeys a uniform law of large numbers for the squared field-distance class. The population criterion has unique minimizer θ_0 .*

THEOREM 15 (Structural consistency). *Under Theorem 14 and Assumption 3, every sequence $\widehat{\theta}_n \in \operatorname{argmin}_{\theta \in \Theta} \widehat{Q}_n(\theta)$ satisfies*

$$(113) \quad \widehat{\theta}_n \rightarrow_p \theta_0.$$

5.3. Geometry identification

Write $h_{\theta}(s) = \operatorname{vech} H_{\mathcal{T},\theta}(s)$ in a fixed local frame and let $G_H(s) = D_{\theta} h_{\theta_0}(s)$. For a positive weighting field $\Omega(s)$ define

$$(114) \quad \mathcal{I}_H = \int G_H(s)' \Omega(s) G_H(s) d\nu(s).$$

PROPOSITION 8 (Local identification from target geometry). *If $\mathcal{I}_H$ is positive definite, θ_0 is locally identified by the target-geometry field. If $\text{rank } \mathcal{I}_H = r < p$, the first-order parameter directions preserved by the geometry field form the r -dimensional range of $\mathcal{I}_H$, while $\ker \mathcal{I}_H$ leaves the field unchanged to first order. Behavioral moments or level-response moments can be stacked with h_θ ; local identification of the joint criterion is equivalent to full column rank of the stacked derivative.*

Operator matching targets parameter combinations that move the state geometry of the declared counterfactual family. Fixed-sieve minimum-distance specification tests follow from the residual between $\widehat{h}^D$ and $h^M(\widehat{\theta})$ after projecting out the model tangent. Response-matching asymptotics and specification testing supply the surrounding minimum-distance framework (Guerron-Quintana, Inoue, and Kilian 2017; Liu, Plagborg-Moller, and Tan 2026).

5.4. First-order distribution

Suppose the field estimator admits an integrated influence representation

$$(115) \quad \sqrt{n} \text{vec}\{\widehat{H}_n(s) - H_0(s)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i(s) + o_p(1)$$

in a function space supporting integration. With $G_\theta(s) = D_\theta \text{vec } H_{\mathcal{T}, \theta_0}(s)$, the field term contributes $-2\lambda \int G_\theta(s)' \xi_i(s) d\nu(s)$ to the score influence function. Combining this term with the behavioral score gives the sandwich covariance.

5.5. Growing sieve and spectral-tail inference

Let $\mathfrak{H}$ be a positive trace-class target operator, $\tau = \text{tr } \mathfrak{H}$, and $\mathfrak{H}_m = \Pi_m \mathfrak{H} \Pi_m$. For a retained dimension d , define

$$(116) \quad \ell_d(\tau, \mathfrak{H}) = \tau - \kappa_d(\mathfrak{H}), \quad \kappa_d(\mathfrak{H}) = \sum_{j=1}^d \lambda_j(\mathfrak{H}).$$

The scalar trace is carried separately from the Hilbert–Schmidt operator so that the stochastic limit can be stated in the Hilbert space $\mathbb{R} \times \mathcal{S}_2^{sa}$.

ASSUMPTION 4 (Joint sieve first stage). *For $m_n \rightarrow \infty$,*

$$(117) \quad \sqrt{n}(\widehat{\tau}_{n, m_n} - \tau_{m_n}, \widehat{\mathfrak{H}}_{n, m_n} - \mathfrak{H}_{m_n}) \Rightarrow (Z_\tau, \mathbb{G})$$

in $\mathbb{R} \times \mathcal{S}_2^{sa}$, and

$$(118) \quad \sqrt{n}\{|\tau_{m_n} - \tau| + \|\mathfrak{H}_{m_n} - \mathfrak{H}\|_2\} \rightarrow 0.$$

A sufficient first-stage condition is an asymptotically linear triangular-array expansion in $\mathbb{R} \times \mathbb{S}_2^{sa}$ with an L^2 -convergent influence function and an $o_p(n^{-1/2})$ remainder.

THEOREM 16 (Spectral-tail limit and numerical directional bootstrap). *Suppose Assumption 4 holds and $\lambda_d(\mathfrak{H}) > 0$. Put $\lambda_* = \lambda_d(\mathfrak{H})$. Let $P_{>}$ project onto eigenvalues strictly above λ_* , let $P_{=}$ project onto the λ_* eigenspace, let $r = \text{rank } P_{>}$, and put $k = d - r$. For a self-adjoint perturbation G , let $\mu_1(G) \geq \dots \geq \mu_q(G)$ be the eigenvalues of $P_{=}GP_{=}$ on the q -dimensional boundary eigenspace. Then*

$$(119) \quad \sqrt{n}\{\widehat{\ell}_{d,n} - \ell_d(\tau, \mathfrak{H})\} \Rightarrow Z_\tau - \text{tr}(P_{>}\mathbb{G}) - \sum_{j=1}^k \mu_j(\mathbb{G}),$$

where $\widehat{\ell}_{d,n} = \widehat{\tau}_{n,m_n} - \kappa_d(\widehat{\mathfrak{H}}_{n,m_n})$. With an eigengap at d , the limit reduces to $Z_\tau - \text{tr}(P_d\mathbb{G})$. Suppose a bootstrap pair $(Z_{\tau,n}^*, \mathbb{G}_n^*)$ consistently estimates the joint limit in (117). If $\epsilon_n \downarrow 0$ and $\sqrt{n}\epsilon_n \rightarrow \infty$, the conditional law of

$$(120) \quad T_n^* = Z_{\tau,n}^* - \frac{\kappa_d(\widehat{\mathfrak{H}}_{n,m_n} + \epsilon_n \mathbb{G}_n^*) - \kappa_d(\widehat{\mathfrak{H}}_{n,m_n})}{\epsilon_n}$$

converges to the law in (119).

The eigenvalue-sum map is a spectral function with a directional derivative determined by the perturbation restricted to each boundary eigenspace (Lewis 1996). The resampling statement applies the directional delta method and its numerical derivative implementation (Fang and Santos 2019; Hong and Li 2018). Sieve inference for increasing-dimensional functionals provides the surrounding approximation framework (Chen and Pouzo 2015). For a declared loss tolerance ϵ , define the sufficient dimension

$$(121) \quad d_\epsilon = \min\{d : \ell_d(\tau, \mathfrak{H}) \leq \epsilon\}.$$

Let $U_{d,n}$ be simultaneous upper confidence bounds satisfying $\Pr\{\ell_d \leq U_{d,n} \ \forall d \leq D\} \rightarrow 1 - \alpha$, and include a terminal dimension D whose tail is zero. Define

$$(122) \quad \widehat{d}_{\epsilon,1-\alpha} = \min\{d : U_{d,n} \leq \epsilon\}.$$

COROLLARY 6 (Confidence-protected sufficient dimension). *Under Theorem 16,*

$$(123) \quad \liminf_{n \rightarrow \infty} \Pr\{\ell_{\widehat{d}_{\epsilon,1-\alpha}} \leq \epsilon\} \geq 1 - \alpha.$$

If d_ϵ is separated by $\ell_{d_\epsilon-1} > \epsilon > \ell_{d_\epsilon}$ and the simultaneous bounds converge uniformly to the tail losses, then $\widehat{d}_{\epsilon,1-\alpha} \rightarrow_p d_\epsilon$. A numerical directional bootstrap obtains simultaneous bounds from the $(1 - \alpha)$ quantile of $\max_d(-T_{d,n}^*)$.

A Monte Carlo uses population eigenvalues (1, 0.50, 0.25, 0.06, 0.06, 0.02) and tolerance $\varepsilon = 0.075$, giving $d_\varepsilon = 5$ with an eigenvalue multiplicity at the adjacent boundary. With sample size 1,000, 500 replications, and 249 multiplier draws, the plug-in rule selects a sufficient dimension in 0.350 of replications. The simultaneous numerical-directional rule selects a sufficient dimension in 0.996 of replications, has mean selected dimension 5.036, and its joint upper bounds cover all population tail losses in 0.966 of replications.

6. Computation

6.1. Planning closure

Counterfactual evaluation uses the quotient in Section 2. Bellman computation uses reward and transition closure. Let $z = b(s)$ be a candidate reduced state.

PROPOSITION 9 (Planning closure). *Suppose that for every action a and policy p in the computational family,*

$$(124) \quad \pi_\theta(a, s, p) = \bar{\pi}_\theta(a, z(s), p)$$

and the pushforward transition satisfies

$$(125) \quad \Pr_\theta\{z(S') \in B \mid s, a, p\} = \bar{P}_\theta(B \mid z(s), a, p)$$

for each measurable B . Then the Bellman value in (1) factors as $V_\theta(s, p) = \bar{V}_\theta(z(s), p)$ and the associated logit conditional choice probabilities factor through $z(s)$.

The proposition connects state compression to the state-equivalence conditions used in MDP model minimization. A target quotient can also be used after solving the full behavioral model to reduce counterfactual response storage and evaluation.

6.2. Multilevel corrections

Let $Y_\ell(\theta)$ denote a structural quantity computed at fidelity level ℓ . It may be a model-implied information operator, a moment, or a score contribution. Define

$$(126) \quad \Delta_\ell Y = Y_\ell - Y_{\ell-1}, \quad \ell \geq 1.$$

The identity

$$(127) \quad \mathbb{E}[Y_L] = \mathbb{E}[Y_0] + \sum_{\ell=1}^L \mathbb{E}[\Delta_\ell Y]$$

supports a coupled estimator with independent sample averages across levels and common random inputs within each fine-coarse pair. Assume the bias, coupled correction variance, and per-draw cost satisfy

$$(128) \quad |\mathbb{E}[Y - Y_L]| \leq c_1 2^{-\alpha L}, \quad V_\ell = \text{Var}(\Delta_\ell Y) \leq c_2 2^{-\beta \ell}, \quad C_\ell \leq c_3 2^{\gamma \ell}.$$

The variance under level sample sizes N_ℓ is $\sum_\ell V_\ell/N_\ell$ and work is $\sum_\ell N_\ell C_\ell$. The cost-minimizing allocation for a variance budget is proportional to

$$(129) \quad N_\ell \propto \sqrt{V_\ell/C_\ell}.$$

PROPOSITION 10 (Multilevel work). *Suppose (128) holds and $\alpha \geq \frac{1}{2} \min\{\beta, \gamma\}$. A coupled multilevel estimator can attain RMSE ε with*

$$(130) \quad \text{Work}_{\text{ML}}(\varepsilon) = \begin{cases} O(\varepsilon^{-2}), & \beta > \gamma, \\ O(\varepsilon^{-2} \log^2 \varepsilon^{-1}), & \beta = \gamma, \\ O(\varepsilon^{-2-(\gamma-\beta)/\alpha}), & \beta < \gamma. \end{cases}$$

The corresponding single-level calculation at the finest required level has work $O(\varepsilon^{-2-\gamma/\alpha})$.

6.3. Multi-index state, angle, and solver refinement

Let $\ell = (\ell_s, \ell_u, \ell_v)$ index state resolution, intervention-angle resolution, and value-function or equilibrium-solver fidelity. In two coordinates,

$$(131) \quad \Delta_s \Delta_u Y_{ij} = Y_{ij} - Y_{i-1,j} - Y_{i,j-1} + Y_{i-1,j-1}.$$

Sparse index sets retain mixed corrections with state-angle-fidelity contributions selected by their estimated target contribution and cost. The multi-index sampling rates follow from mixed-difference bias, variance, and cost exponents as in Haji-Ali, Nobile, and Tempone (2016). For a current rank- d state projector, a candidate target view can be ranked by

$$(132) \quad G(t; s) = \text{tr} \left[(I - P_d(s)) A_t(s)' W_t A_t(s) \right].$$

A computational acquisition rule can integrate (132) against a state uncertainty weight and choose the next target angle or fidelity block from the feasible menu.

6.4. Equivalence to an exact numerical criterion

Let Q_n^E be the sample structural criterion computed by an exact numerical solver and Q_n^{ML} the criterion computed by the multilevel scheme.

ASSUMPTION 5 (Numerical score and curvature). *The exact estimator is regular at an interior parameter θ_0 : $\sqrt{n}\nabla Q_n^E(\theta_0) \Rightarrow N(0, \Omega)$ and $\sup_{\theta \in N} \|\nabla^2 Q_n^E(\theta) - J\| \rightarrow_p 0$ on a neighborhood N , where J is nonsingular. The numerical criterion satisfies*

$$(133) \quad \sup_{\theta \in N} \|\nabla Q_n^{ML}(\theta) - \nabla Q_n^E(\theta)\| = o_p(n^{-1/2})$$

and

$$(134) \quad \sup_{\theta \in N} \|\nabla^2 Q_n^{ML}(\theta) - \nabla^2 Q_n^E(\theta)\| = o_p(1).$$

THEOREM 17 (Numerical equivalence). *Let $\widehat{\theta}_E$ and $\widehat{\theta}_{ML}$ be local minimizers of Q_n^E and Q_n^{ML} in N . Under Assumption 5,*

$$(135) \quad \sqrt{n}(\widehat{\theta}_{ML} - \widehat{\theta}_E) = o_p(1)$$

and

$$(136) \quad \sqrt{n}(\widehat{\theta}_{ML} - \theta_0) \Rightarrow N(0, J^{-1}\Omega J^{-1}).$$

7. Diagnostic structural experiments

The numerical diagnostics examine the representation, estimation, design, and computation steps separately. Appendix B gives their construction. The target-dependent DDC sets the dominant baseline value-function directions apart from the finite-counterfactual directions. Figure 1 reports held-out response error as the retained dimension varies. The exact-closure exercise separates target sufficiency from Bellman sufficiency. In that design the reward and pushforward transition law factor through two count states, so the Bellman problem closes exactly. The scattered and boundary-spectrum exercises evaluate the first-stage and inferential conditions used by Theorems 14 and 16. The multi-index exercise evaluates the numerical decomposition in Section 6.

8. Model-based distributional state in general equilibrium

The quantitative illustration is an incomplete-markets transition economy with one asset and three idiosyncratic income states. Households have CRRA utility with discount factor 0.94 and risk aversion 2. Next-period asset choices lie on 15 points over $[0, 20]$ and receive

TABLE 1. Diagnostics for the structural representation

Exercise	Object	Result
Exchangeable-state DDC	Planning closure at $K = 14$	245,760 Bellman states map to 960 count states; mapped values and choice probabilities agree to numerical precision.
Target-dependent DDC	Two-coordinate held-out response fit	Counterfactual coordinates: 0.225 response SD; baseline-value coordinates: 1.002 response SD.
Scattered response field	Cross-fitted local quadratic patches	Projector error falls as anchors increase; same-sample squares retain the quadratic noise component.
Intervention allocation	Goal-oriented posterior target loss	Added policy views reduce the residual information directions selected by the target criterion.
State-angle refinement	Multi-index mixed corrections	State and angle resolution enter separate telescoping axes.
Loss-controlled dimension	$\varepsilon = 0.075$ with an adjacent eigenvalue multiplicity	Plug-in sufficiency: 0.350; simultaneous numerical-directional rule: 0.996; mean protected dimension: 5.036.

type-I extreme-value smoothing with scale 0.8. Income levels are (0.5, 1, 1.5) with transition matrix

$$(137) \quad P_y = \begin{pmatrix} 0.90 & 0.09 & 0.01 \\ 0.05 & 0.90 & 0.05 \\ 0.01 & 0.09 & 0.90 \end{pmatrix}.$$

Production is $Y = K^{0.36}$ with depreciation 0.08. Competitive factor prices are

$$(138) \quad r(K) = 0.36K^{-0.64} - 0.08, \quad w(K) = 0.64K^{0.36}.$$

The stationary equilibrium has $K = 4.4804$, $r = 0.0579$, and $w = 1.0981$. Each transition path is solved by backward household recursion and forward distribution propagation with market clearing for aggregate capital. The household distribution has 45 mass points. The supplied representation records total mass, aggregate assets, and the masses in the first two income states. The third income mass follows from total mass. In Fisher score coordinates the residual distribution space has dimension 41. The declared counterfactual family contains two balanced fiscal packages. A capital-income tax begins at 0.25 and decays geometrically at rate 0.5 with contemporaneous lump-sum rebates. A labor-income tax begins at 0.30 and has the same decay and rebate rule. The target vector contains capital, consumption, output, and the interest rate at horizons 0, 1, 2, 4, and 8 for both packages, giving 40 target components. The supplied map, fiscal packages, horizons, outcome nor-

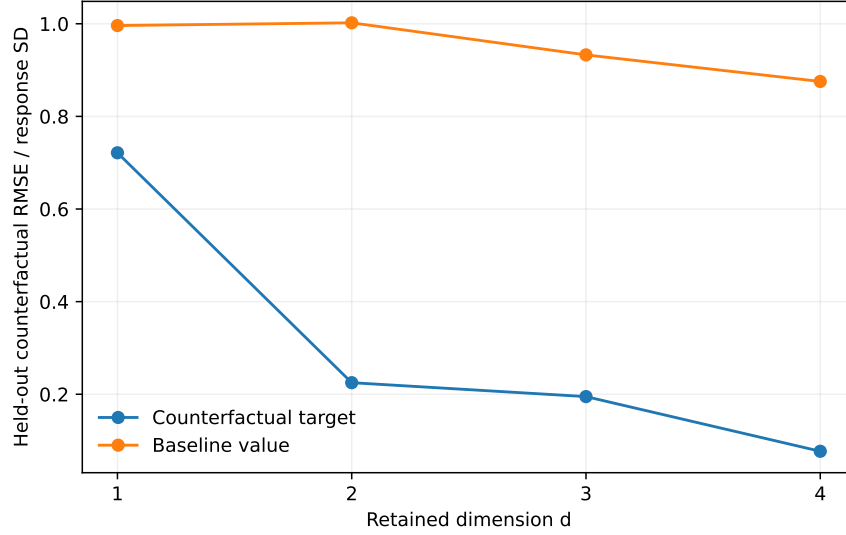


FIGURE 1. Counterfactual response error by retained dimension

Each representation is estimated on a training sample of states and evaluated on held-out finite policy responses. The target coordinates use the counterfactual derivative field. The baseline-value coordinates use the gradient field of the baseline value function. Errors are divided by the held-out response standard deviation.

TABLE 2. Distributional representation and fiscal-counterfactual loss

Coordinates d	Counterfactual loss share	Moment-dictionary loss share	Cumulative target share
1	0.274	0.526	0.726
2	0.031	0.439	0.969
3	0.012	0.376	0.988
4	0.005	0.338	0.995
5	0.001	0.304	0.999
6	0.0007	0.293	0.9993

Losses are divided by the zero-coordinate target loss. The moment dictionary contains powers of assets, asset-income interactions, and wealth-income cells. Each moment step selects the remaining dictionary element that gives the largest target-loss reduction.

malization, state metric, and target metric are common across the scattered-state exercise. The response field is evaluated along 46 scattered Fisher-score directions that preserve the supplied representation. Central finite differences produce directional derivatives of the nonlinear general-equilibrium transition paths. A least-squares reconstruction of the derivative map has RMSE $1.23e - 08$, and the standard deviation of the directional response derivatives is $4.45e - 03$. The resulting target spectrum has cumulative shares 0.726, 0.969, and 0.988 after one, two, and three coordinates. The comparison has a measurement interpretation. Under the unrestricted local linear class in Theorem 12, the target eigen-directions minimize residual derivative loss. The moment dictionary restricts

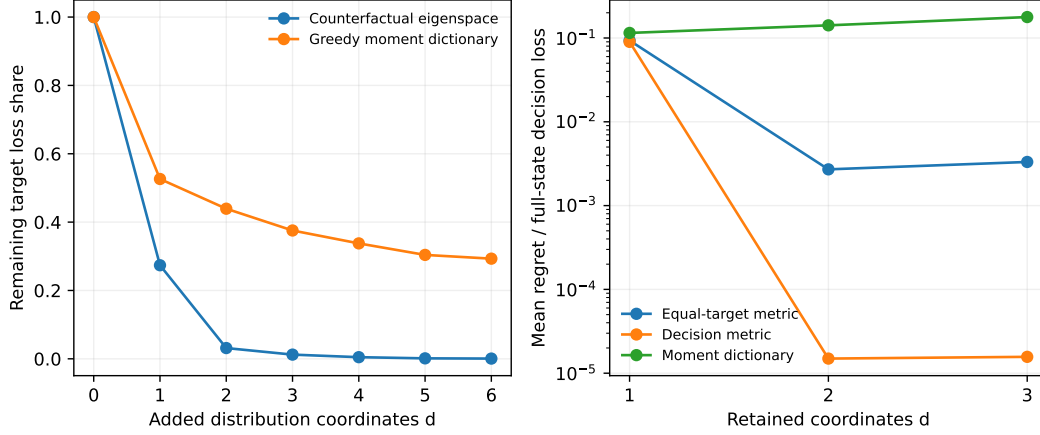


FIGURE 2. Distributional coordinates in the general-equilibrium transition economy

Left: remaining fiscal-counterfactual target loss at each dimension. Right: mean held-out policy regret relative to the mean full-state decision loss for the equal-target metric, the decision metric, and the moment dictionary. The decision-metric coordinates use the local regret metric in Theorem 6.

added measurements to its listed economic moments. The two columns therefore answer different admissible-class questions. Table 2 reports the resulting class-conditional frontier for these two admissible classes. The distributional state can also be selected for a downstream policy decision. Let $G(s)$ be the 20×2 matrix whose columns are the two package-response paths. The planner chooses package weights $a \in \mathbb{R}^2$ to minimize

$$(139) \quad \frac{1}{2} \|d_0 + G(s)a\|^2 + \frac{\eta}{2} \|a\|^2, \quad \eta = 2 \times 10^{-7},$$

where d_0 is fixed at one half of the negative sum of the two baseline package responses. The Hessian calculation in Theorem 6 gives a target metric on $\text{vec } G$. Three representations are estimated on a training set of scattered distributions: the equal-target metric, the decision metric, and the greedy moment dictionary. The response surface is predicted on held-out distributions and the planner resolves (139) using the predicted response. At $d = 2$, the decision-metric representation has policy-weight RMSE 1.08e-04, the equal-target representation has 1.41e-03, and the moment dictionary has 5.13e-03. Their mean regret relative to the mean full-state decision loss is 1.49e-05, 0.003, and 0.142. Figure 2 reports the target-loss and decision comparisons. The distributional derivative calculation is the finite-state counterpart of the Hilbert-space operator in Section 2.5. Sequence-space methods compute equilibrium derivatives in heterogeneous-agent models (Auclert et al. 2021), and projection and parametric-distribution methods provide state representations (Reiter 2009; Winberry 2018). The calculation here starts from the supplied aggregate representation and measures the distributional directions required by the declared fiscal transition paths. Every response in this section is generated by the calibrated model. The

evidence establishes internal approximation for the stated policies, metric, perturbation law, and coordinate classes. Empirical validation requires an independently recovered intervention-response field evaluated under the same contract.

9. Finite-graph coordinate selection in a fixed KW97 model

The local spectrum supplies a relaxation whose global performance depends on the feature class, perturbation distribution, and decoder. A fixed-model KW97 exercise applies this distinction in a dynamic occupational-choice model. The exercise solves five policies on aligned state spaces under two distributed parameterizations. Each parameterization contains 6,520 core states and four latent types. The target vector contains 55 state-level policy responses: EMAX, five choice probabilities, and five alternative-specific mean values under each policy. Core states enter a 70/30 training-test split, with all latent types of a core state assigned to the same fold. The primary supplied representation records period, schooling, total civilian experience, and lagged-choice indicators. Its fiber contains the civilian experience mix, military experience, and three latent-type contrasts. Every method carries the same supplied representation. Counterfactual-fiber, baseline-value-fiber, and unrestricted-output methods receive three added coordinates and use the same quadratic ridge decoder. The full-state row uses all five omitted coordinates. Table 3 averages held-out target RMSE across two derivative-estimation runs.

TABLE 3. KW97 fixed-model held-out representation stress test

Representation	Distributed respy	Distributed original
Counterfactual fiber, $d = 3$	0.7356	0.7011
Baseline-value fiber, $d = 3$	0.6758	0.6844
Unrestricted output, $d = 3$	0.7121	0.7065
Full state, $d = 5$	0.6488	0.6435

Entries are mean held-out RMSE for the common 55-component policy-response vector. Representations use the same supplied map, state split, training-standardized metric, decoder, and simulation design.

Across the supplied-representation specifications, the counterfactual-fiber coordinates remain above the best equal-budget competitor under the stricter cases. In the primary specification, the first three counterfactual directions retain about 78–83 percent of positive derivative trace, yet their global held-out RMSE exceeds the baseline-value comparator by 8.9 percent under the respy parameterization and 2.4 percent under the original parameterization. All methods preserve the ranking of mean EMAX policy responses. This result realizes the separation in Theorem 9: derivative concentration leaves fiber-frequency slack and global decoding risk undetermined.

The finite-graph exercise implements Proposition 5 on the same split. Two training folds produce out-of-fold residuals from the common quadratic ridge decoder. Exact cells of the primary supplied representation fix period, schooling, total civilian experience, and lagged choice. Within each cell, a complete reversible graph assigns inverse-distance weights in the candidate’s added coordinates; the bandwidth is the median positive within-cell distance. The graph certificate, candidate menu, target normalization, and decoder use training states. Held-out targets enter only the final comparison. The menu contains the counterfactual-fiber spectrum, baseline-value spectrum, and conventional coordinates at budgets $d = 1, 2, 3$.

At $d = 3$, certificate minimization selects the baseline-value spectrum under both parameterizations and both derivative-estimation runs. That candidate is also the held-out risk minimizer in all four comparisons. Relative to the counterfactual-fiber spectrum, it lowers held-out target RMSE by 1.80–8.21 percent. A paired bootstrap over the 1,956 held-out core states gives intervals that exclude zero in every comparison. Table 4 reports the selection calculation. The repeated-split audit fixes the 55-target scale and active

TABLE 4. Training-selected graph certificate and held-out KW97 risk, $d = 3$

Model / run	$\widehat{\mathcal{C}}_G(\text{CF})$	$\widehat{\mathcal{C}}_G(\text{selected})$	RMSE(CF)	RMSE(selected)	Reduction [95% CI]
respy / 1	0.896	0.811	0.7262	0.6677	8.06 [7.37, 8.78]
respy / 2	1.003	0.795	0.7450	0.6839	8.21 [7.56, 8.93]
original / 1	0.842	0.751	0.7065	0.6856	2.95 [2.60, 3.34]
original / 2	0.845	0.825	0.6957	0.6832	1.80 [1.53, 2.09]

CF denotes the local counterfactual-fiber spectrum. The selected candidate is the baseline-value spectrum in every row. Certificates use two-fold out-of-fold training residuals. Intervals are percentile intervals from 1,999 paired core-state bootstrap draws of the held-out RMSE reduction, in percent.

mask at the original design, then reruns the core-state partition, state standardization, local operators, ridge selection, graph construction, candidate selection, and held-out evaluation under 20 split seeds. Table 5 separates the direct risk score from the theorem-linked graph certificate. Minimum two-fold out-of-fold decoder risk matches the held-out oracle in all 80 model–run–split comparisons. It selects the baseline-value family in 75 comparisons and the counterfactual-fiber family in five. The median RMSE gain relative to the counterfactual-fiber family is 4.29 percent. The four-comparison mean gain is positive in every split and ranges from 2.76 to 7.67 percent. Paired 1,999-draw intervals exclude zero in all 75 comparisons where the direct selector chooses the baseline-value family. The graph certificate matches the held-out oracle in 63 of 80 comparisons and remains the diagnostic tied to the population certificate bound. Across the twelve $d = 3$ candidate–model–run observations in the original split, the Spearman correlation between the graph certificate and held-out RMSE is 0.926; the correlation between retained counterfactual

TABLE 5. Repeated-split KW97 coordinate selection, $d = 3$

Training score	Oracle	BV / CF	Median gain	Split means	Positive CIs
Out-of-fold decoder risk	80 / 80	75 / 5	4.29	[2.76, 7.67]	75 / 80
Graph certificate	63 / 80	62 / 18	4.16	[0.61, 7.67]	60 / 80

The 80 comparisons comprise 20 core-state splits, two fixed-model parameterizations, and two derivative-estimation runs. Gains and ranges are percentage reductions in held-out target RMSE relative to the local counterfactual-fiber spectrum. Positive intervals count comparisons whose paired percentile interval excludes zero.

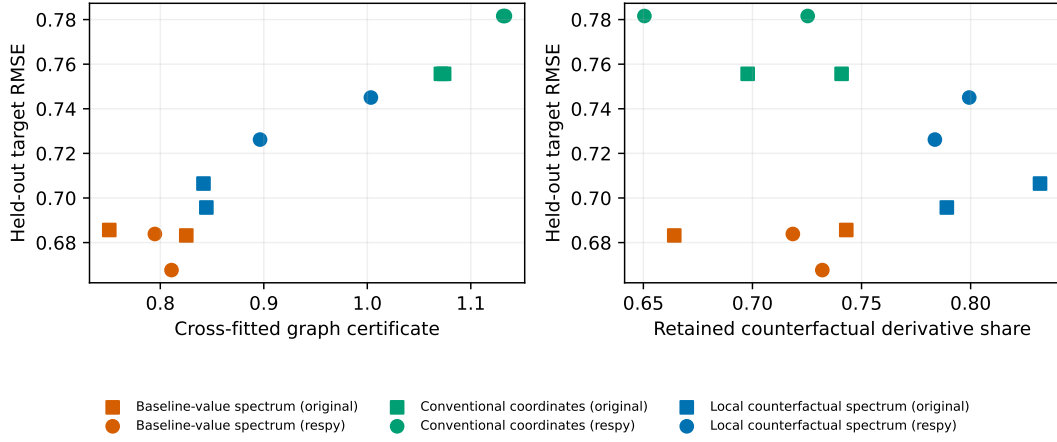


FIGURE 3. Generator certificate, derivative share, and held-out target risk

Each point is one candidate under one parameterization and estimation run at $d = 3$. Circles denote the distributed respy parameterization and squares the distributed original parameterization. The left panel compares the training-sample graph certificate with held-out RMSE. The right panel compares retained counterfactual derivative share with the same held-out outcome.

derivative share and held-out RMSE is -0.112 . Across all budgets in that split, certificate minimization selects the held-out best candidate in six of twelve model-run-budget comparisons, while direct out-of-fold risk selection does so in twelve of twelve. The 20-split audit shows how the two training scores separate: direct risk reproduces the three-coordinate oracle throughout the audit, while the graph certificate provides the theorem-linked upper-bound diagnostic. The policies and targets are generated within the structural model. The exercise evaluates the coordinate-selection mechanism under a fixed model; causal response validation comes from external evidence.

10. Conclusion

A supplied representation and a declared counterfactual family define an exact quotient of structural states. Within each supplied-state fiber, the local target operator provides a smooth relaxation. Its rank gives the local repair count, and its spectral tail gives the

leading decoder risk in a shrinking isotropic experiment. Actual coordinates must come from a declared admissible class. Their level sets impose integrability, their conditional distributions determine Poincaré constants, and disconnected components can add target variation that within-stratum derivatives do not capture. Global risk also depends on fiber geometry and the frequency content of the target. Under the weighted generator, the fiber certificate measures the higher-frequency component of the conditional target residual. The feasible-coordinate bound then compares certificate-based selection with the best global risk in a declared finite-complexity class. Its reported form requires uniform estimates of conditional gaps, density scores, target derivatives, and target second derivatives. In the heterogeneous-agent calculation, the fiscal target family is concentrated under an unrestricted local measurement class. In the KW97 exercise, direct out-of-fold risk selects the held-out three-coordinate oracle in all 80 repeated-split comparisons, with a 4.29 percent median RMSE gain over the local spectrum. The graph certificate matches 63 of 80 held-out oracles and records the finite-state counterpart of the population bound. These exercises use model-generated response fields under declared target and coordinate contracts. Validation against data requires response derivatives recovered from independent intervention variation.

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